Gene Expression Prediction in Single Cells via Data-Driven Discrete Dynamical Systems

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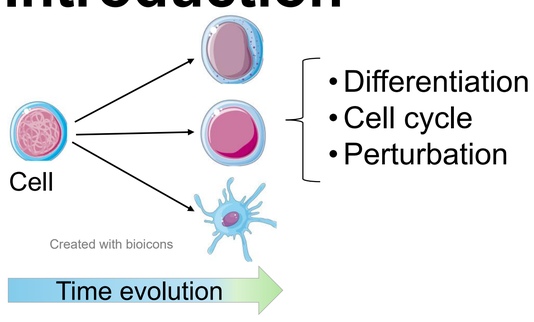
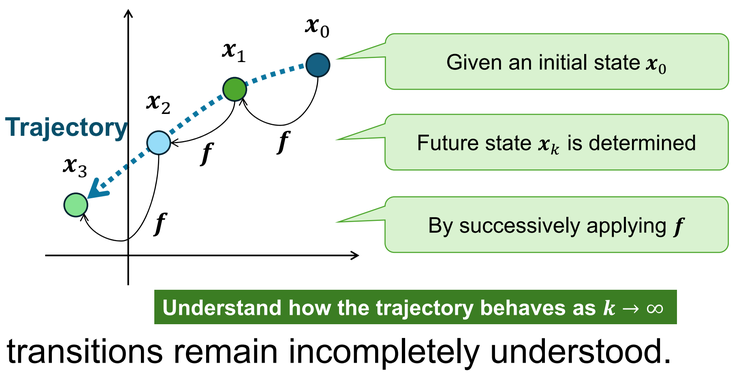
Introduction



Discrete dynamical system: $\boldsymbol{x}_{k+1} = \boldsymbol{f}(\boldsymbol{x}_k)$

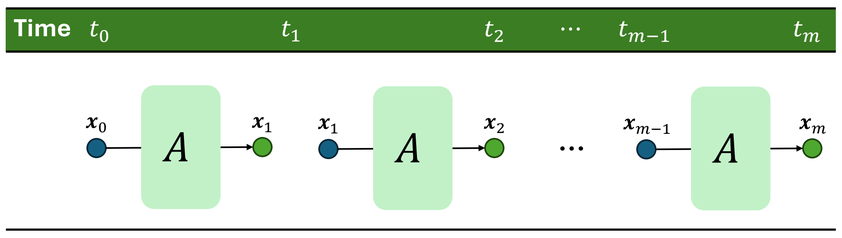


Governing principles of cell state transitions remain incompletely understood.

Aim: **Elucidate the rules governing cell state transitions**

- No governing equations assumed in advance
- Use snapshot pairs $\{(\boldsymbol{x}_k, \boldsymbol{x}_{k+1}) \mid k = 0, \dots, m-1\}$ to infer a dynamical system

Data-driven linear discrete dynamical system: $\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k$

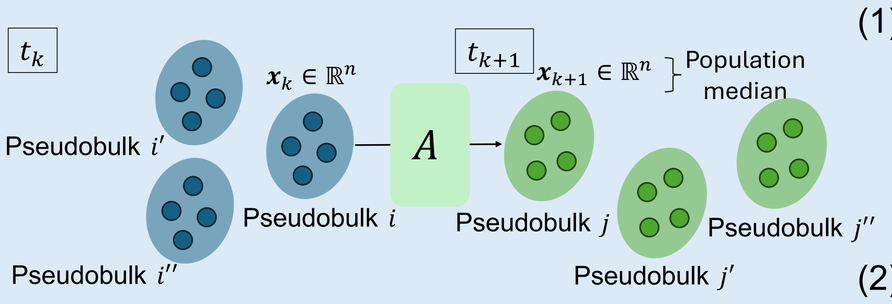


Methods

1. Create pseudobulk snapshot pairs

Input: time-series scRNA-seq datasets

(1) Perform clustering of each dataset



(2) Find mapping from i to j by solving the entropic regularization **optimal transport** problem

Output: snapshot pairs for each pseudobulk:

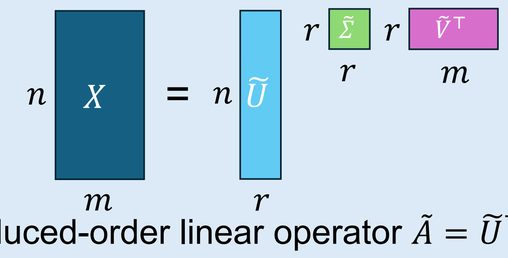
$$X = (\boldsymbol{x}_0, \boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}) \in \mathbb{R}^{n \times m}$$
$$X' = (\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_m) \in \mathbb{R}^{n \times m}$$

2. Perform dynamic mode decomposition (DMD) [1,2]

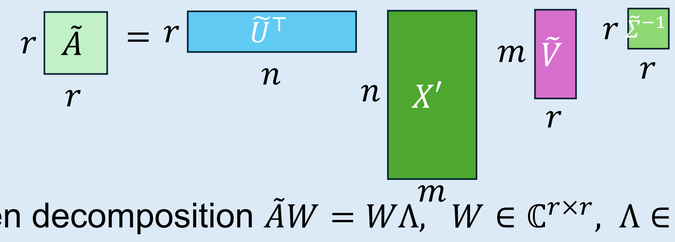
Input: snapshot pairs $X = (\boldsymbol{x}_0, \boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}) \in \mathbb{R}^{n \times m}$, $X' = (\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_m) \in \mathbb{R}^{n \times m}$, rank $r \in \mathbb{N}$

Aim: **Find a matrix $A \in \mathbb{R}^{n \times n}$ such that $X' \approx AX$** ➡ **Compress the data to size r ($\because n$ is large)**

(1) Compute a truncated **singular value decomposition (SVD)** of $X \approx \tilde{U}\tilde{\Sigma}\tilde{V}^\top$, $\tilde{U} \in \mathbb{R}^{n \times r}$, $\tilde{\Sigma} \in \mathbb{R}^{r \times r}$, $\tilde{V} \in \mathbb{R}^{m \times r}$



(2) Compute the reduced-order linear operator $\tilde{A} = \tilde{U}^\top X' \tilde{V} \tilde{\Sigma}^{-1} \in \mathbb{R}^{r \times r}$

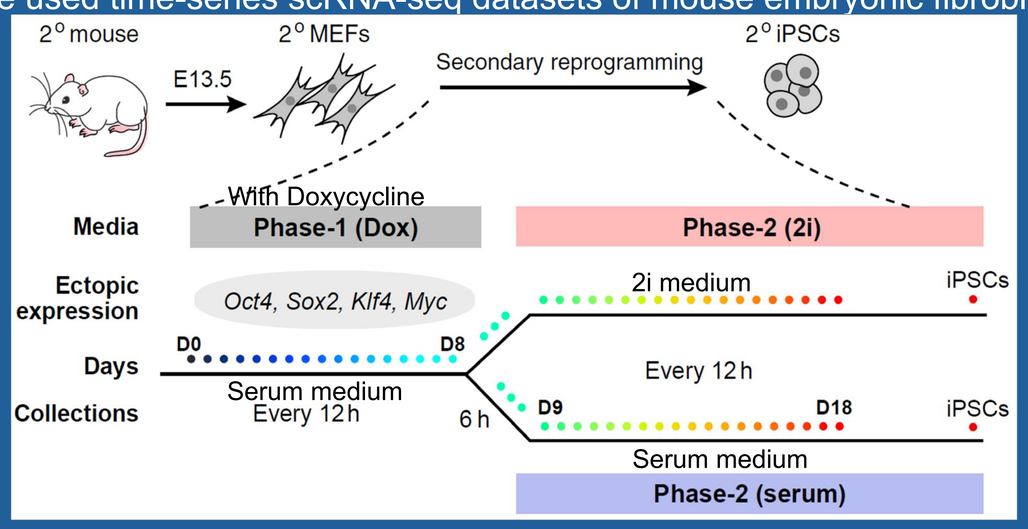


(3) Compute the eigen decomposition $\tilde{A}W = W\Lambda$, $W \in \mathbb{C}^{r \times r}$, $\Lambda \in \mathbb{C}^{r \times r}$

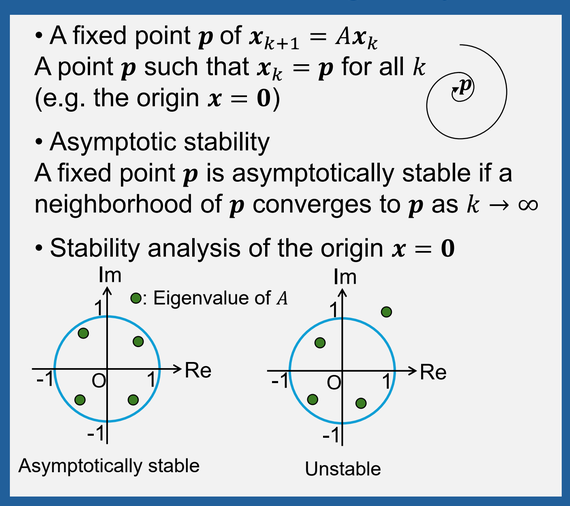
Output: the reduced-order linear operator $\tilde{A} \in \mathbb{R}^{r \times r}$, the eigenvalues $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \dots, \lambda_r) \in \mathbb{C}^{r \times r}$

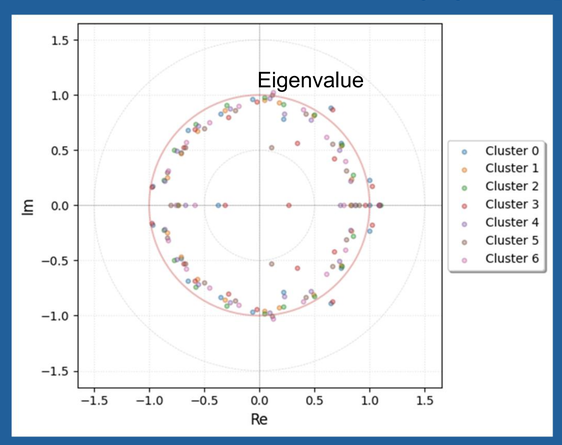
Results

We used time-series scRNA-seq datasets of mouse embryonic fibroblasts cultured in 2i medium following 8 days of culture in serum medium with doxycycline [3].

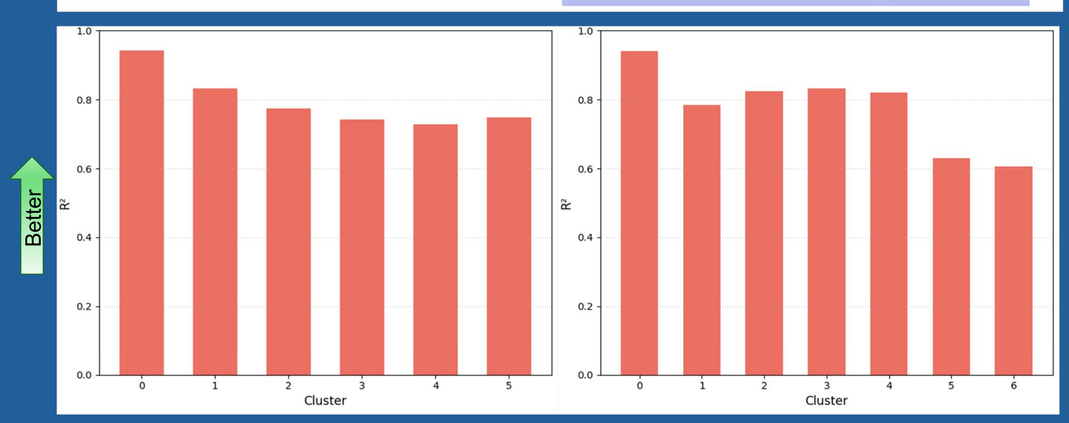


- A fixed point $\boldsymbol{p}$ of $\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k$
  A point $\boldsymbol{p}$ such that $\boldsymbol{x}_k = \boldsymbol{p}$ for all k (e.g. the origin $\boldsymbol{x} = \boldsymbol{0}$)
- Asymptotic stability
  A fixed point $\boldsymbol{p}$ is asymptotically stable if a neighborhood of $\boldsymbol{p}$ converges to $\boldsymbol{p}$ as $k \to \infty$
- Stability analysis of the origin $\boldsymbol{x} = \boldsymbol{0}$



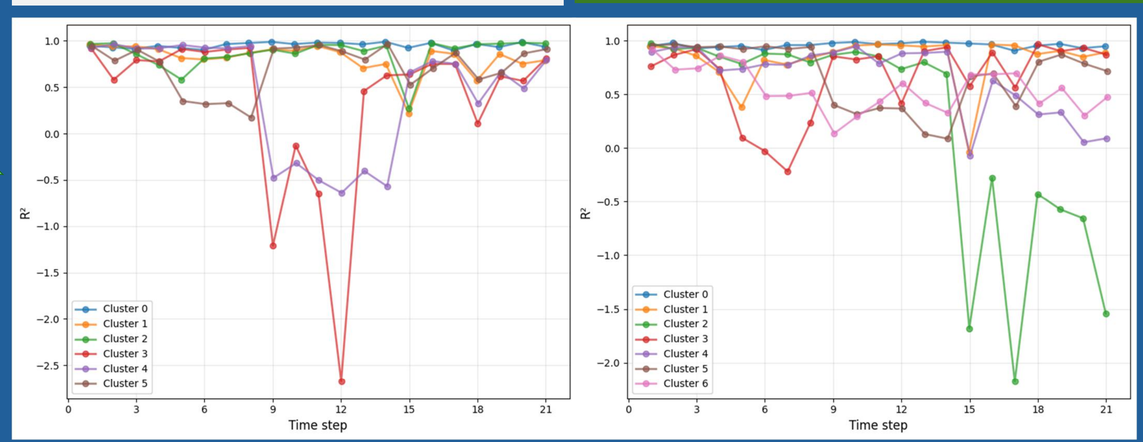


States with low expression in multiple cell types are unstable.



Train on replicate C1; test on replicate C2. Train on replicate C1; test on replicate C2.
R^2: coefficient on determination (agreement between predictions and true values)

The one-step-ahead predictions work reasonably well.



Train on replicate C1; test on replicate C2. Train on replicate C1; test on replicate C2.

Are the predictions for $m = 2, 3, \dots$ steps ahead not very good ?

Discussion

- The DMD-based approach can predict near-future gene expression: useful stability analysis and good predictive performance.
- Adopting truncated SVD for dimensionality reduction provides effective noise reduction, particularly for sparse scRNA-seq datasets.
- Future work
  - Reduce the noise in observations within a linear framework by considering both forward and backward dynamics.
  - Extend linear dynamical systems to handle nonlinearity using extended DMD or Koopman operators.

References

[1] Tu et al., On dynamic mode decomposition: theory and applications, *J Comput Dyn*, 1, 391–421, 2014.
[2] Colbrook, The multiverse of dynamic mode decomposition algorithms, *arXiv*, 2023.
[3] Schiebinger et al., Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming, *Cell*, 176, 928–943, 2019.